

Introduction

Electricity’s antimicrobial properties have shown promise in fighting antibiotic resistant bacteria. Treating bacteria for just 30 minutes with **low-current electricity** **damages the cell membrane** and biofilm, which leads to an inability to control its interior contents (Rogers 20). **Copper ions** are also highly deadly to the cell. It should be possible to design a doorknob that first **opens the cell** by damaging its cell membrane to contamination then kills them using copper ions. This mechanism is likely useful for **reducing sepsis deaths** in hospital and damage bacterial biofilm before it has a chance to grow.

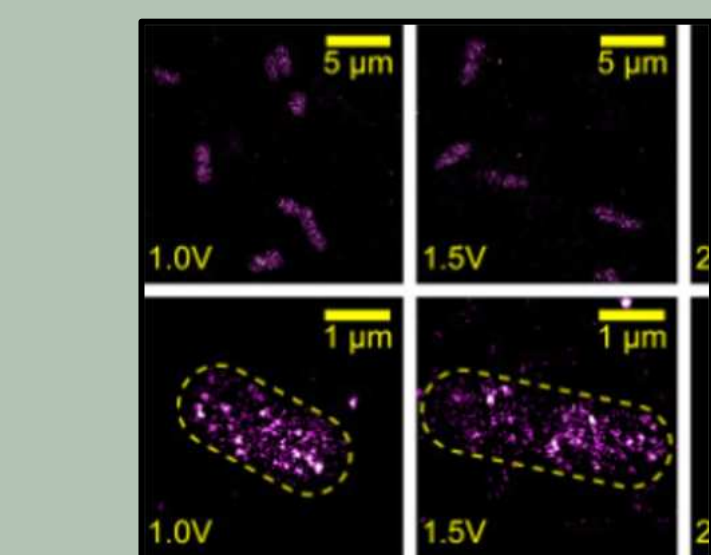


Fig 1. Cell membranes leaking ions due to electrical damage (Vestby, 2020)

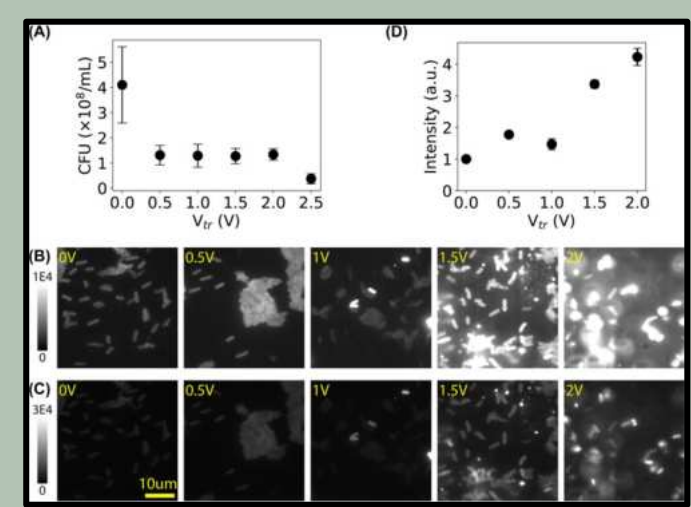


Fig 2. Bacterial death over time using just low-current electricity (Krishnamurthi, 2020)

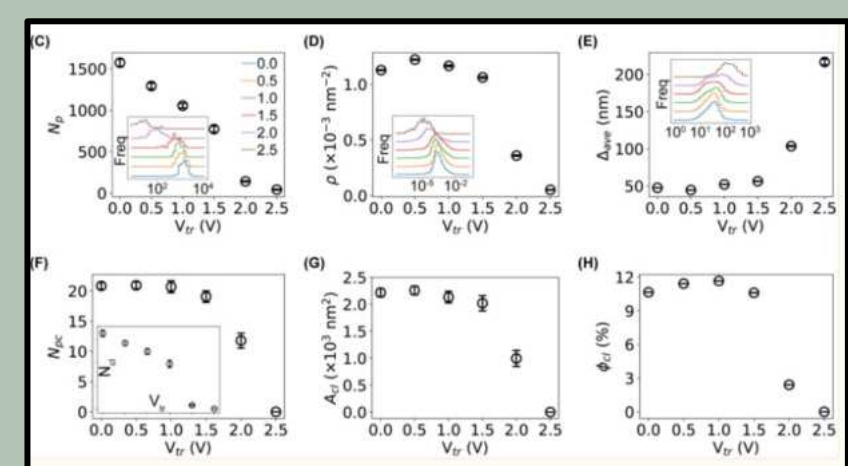


Fig 3. Presence of bacterial proteins over time using low-current electricity (Krishnamurthi, 2020)

Purpose

Bacterial biofilm contributes to the development of cancer and is part of most chronic infections, and doorknobs are one of the largest disease vectors (Vestby 20). This biofilm contributes to **antibiotic resistance**. Even if treatments work initially, as the disease reproduces, it adapts. Current solutions include brass doorknobs and cleaning fluids, both of which have issues like tarnishing or manual application.

As such, a device that uses a physical method to kill bacteria automatically without tarnishing would help mitigate the effects of bacterial biofilms in hospitals greatly.

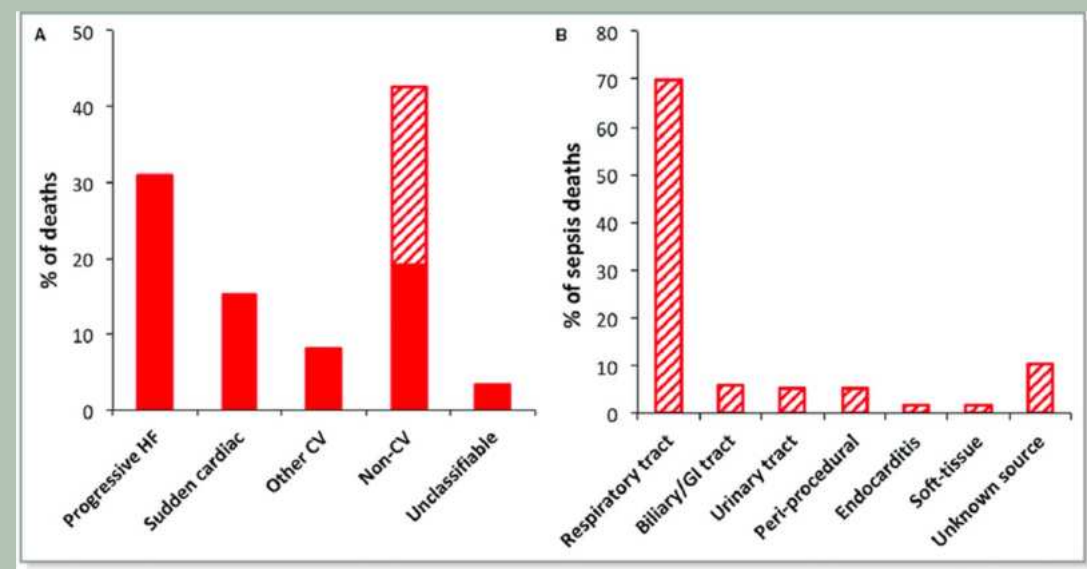


Fig 4. Sepsis deaths compared to other common causes. Translucent part is sepsis (Walker, 2018)

Analyzing the Efficacy of Electricity in Improving the Penetration of Germicidal Material into Antibiotic Resistant Bacteria

Methodology



- Prepare wiring diagram for both test and experimental electricity model
- Connect copper bar to 3d printed chassis
- Run power through the circuit to ensure that the test will work
- Construct all necessary apparatuses

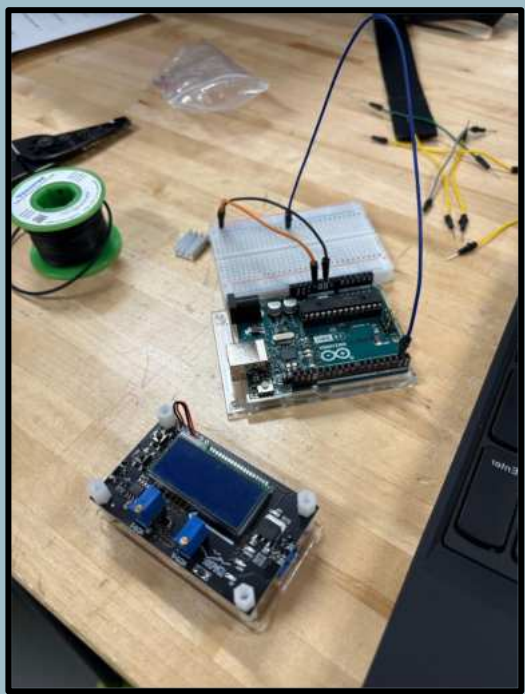


Fig 5. Basic wiring setup

- Rehydrate bacteria
- Mark petri dishes and swab them with E.Coli
- Set up petri dishes for the experiment
- Make sure there is enough bacteria for experimentation
- Ready electricity module

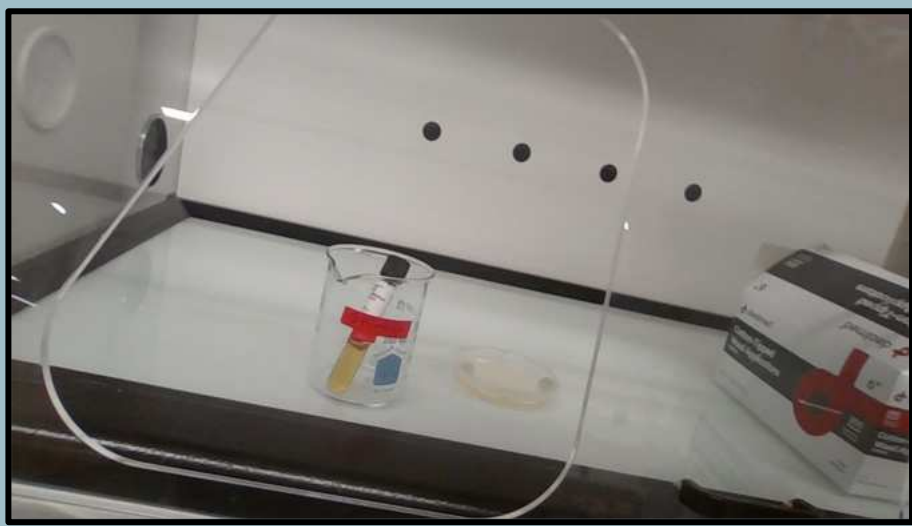


Fig 6. Swabbing plate with E.Coli for control

- Run copper, copper with electricity, and control tests
- Swab residual bacteria from each
- Regrow bacteria in order to create new samples
- Culture each solution in a fresh agar plate, then count colony forming units

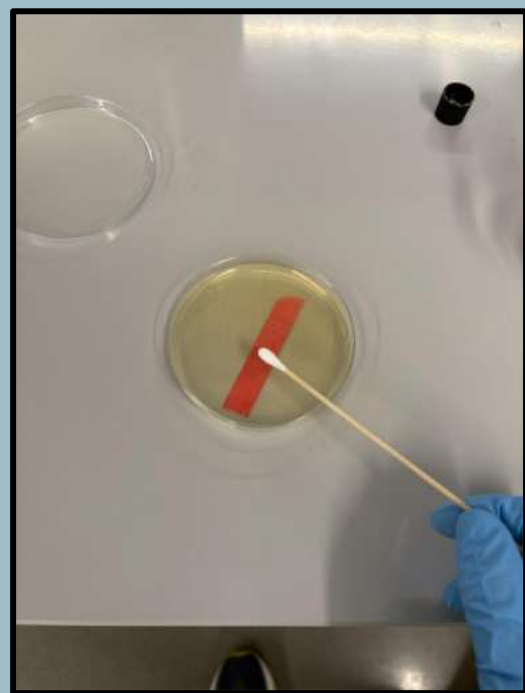


Fig 7. Initial bacteria culturing and rehydration

Conclusion

As can be seen in the graph image, the electrified copper’s line is **consistently below** the non-electrified copper’s line. As well as this, it uniquely goes to 0 colony forming units after 2 hours. The **control data was non-useful** for the purpose of analysis, so it was not included in this graph. This was because they always grew lawns, simply ensuring that the experiment didn’t have unintended disinfectants. From the data, It is clear that this approach is **more effective** at killing E.Coli K-12 than just electricity. The electrified copper was more effective at all times, and included the only trial able to achieve **full sterilization**. Overall, this means that the two mechanisms **work together** in tandem. We can safely **reject the null hypothesis** and conclude that **electrified copper is a highly effective sterilizing agent**. Previous data collected from Vestby 2020 and Krishnamurthi 2020 using only low-current electricity as a disinfectant found a significant decrease in, but not a complete cessation of bacterial activity. As such, because copper and electricity together reduced bacterial growth to nothing, it is clear that the two work together well.

Results



Fig 8. Results of the 30 minute electricity only test

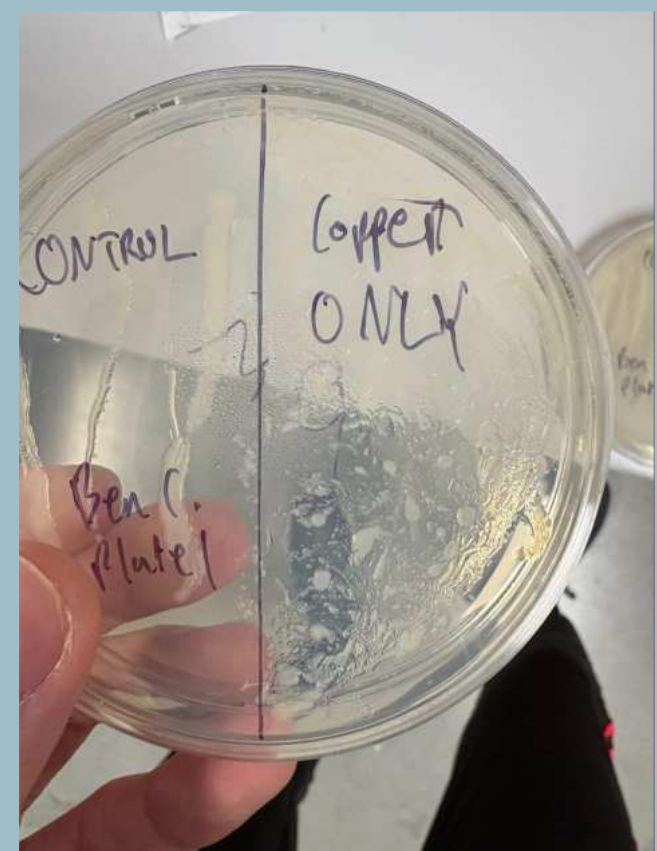


Fig 11. 15 minute copper test. Note clear growth even on the copper side

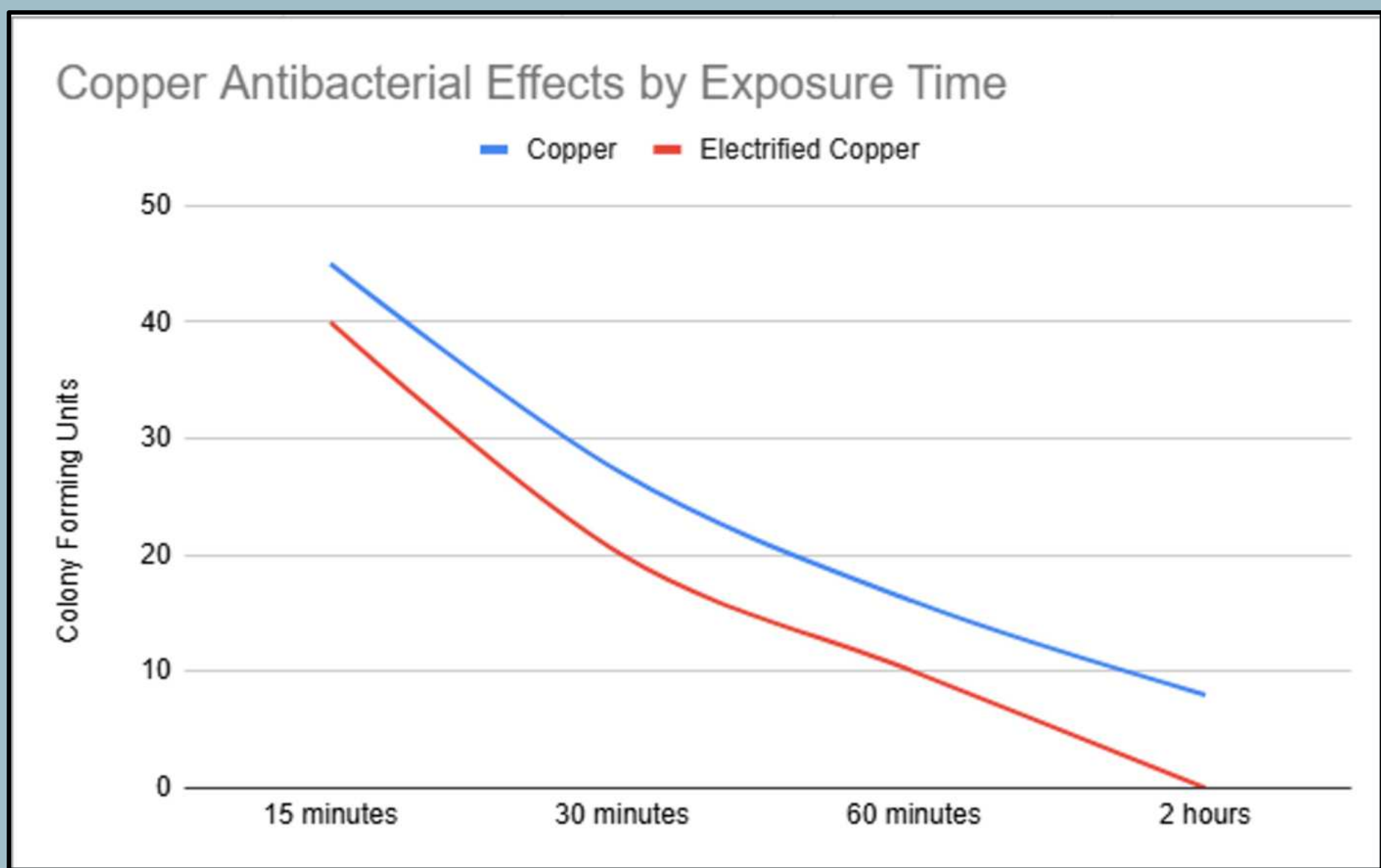


Fig.9 Line graph of copper vs electrified copper

	15 minutes	30 minutes	60 minutes	2 hours
Copper	45	27	16	8
Electrified Copper	40	20	10	0
Control	Lawn	Lawn	Lawn	Lawn

Fig.12 Data table of results + comparison

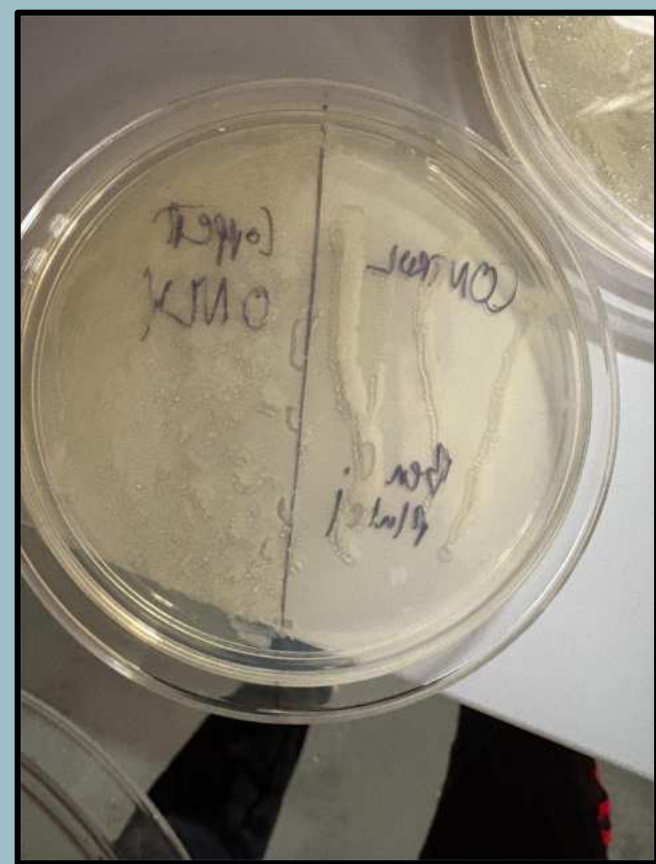


Fig 10. 30 minute copper test. Slightly less growth than 15, but very noticeable



Fig 13. 2 hour trial with electricity showing no growth across the entire dish. Smudges are from condensation

Extensions

One purpose already mentioned for these results would be to use for **sterilization in a hospital setting**. These would be effective for increasing safety or high-risk patients and reducing labor costs. As well as this, the mechanism is **cheap**, meaning it could be widely implemented without issue. Further tests should include more trials.

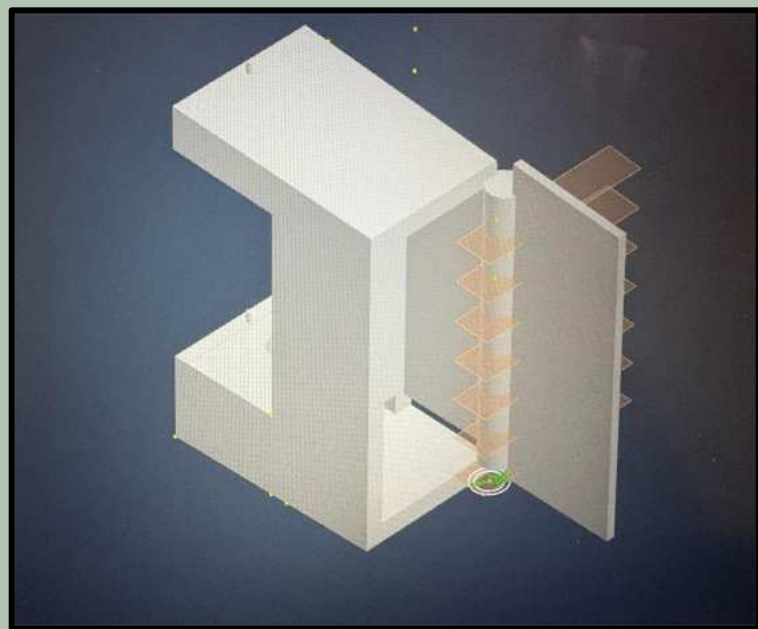


Fig 14. 3D model of what a design utilizing this might look like

References

Salah, Intisar, et al. "Copper as an Antimicrobial Agent: Recent Advances." *RSC Advances*, U.S. National Library of Medicine, 19 May 2021, [pmc.ncbi.nlm.nih.gov/articles/PMC9033467/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC9033467/). Krishnamurthi, V. R., Rogers, A., Peifer, J., Niyonshtu, I. I., Chen, J., & Wang, Y. (2020, August 3). *Microampere electric current causes bacterial membrane damage and two-way leakage in a short period of time*. Applied and environmental microbiology. <https://pubmed.ncbi.nlm.nih.gov/articles/PMC7414946/>

Vestby, L. K., Gronseth, T., Simm, R., & Nesse, L. L. (2020, February 3). *Bacterial biofilm and its role in the pathogenesis of disease*. Antibiotics (Basel, Switzerland). <https://pubmed.ncbi.nlm.nih.gov/articles/PMC7167820/>

Ventola, C Lee. "The Antibiotic Resistance Crisis: Part 1: Causes and Threats." *P & T: A Peer-Reviewed Journal for Formulary Management*, U.S. National Library of Medicine, Apr. 2015, [pmc.ncbi.nlm.nih.gov/articles/PMC4378521/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC4378521/).

Walker, Andrew. "Prevalence and Predictors of Sepsis Death in Patients with Chronic Heart Failure and reduced Left Ventricular Ejection Fraction." *Journal of the American Heart Association*, U.S. National Library of Medicine, Oct. 2018, pubmed.ncbi.nlm.nih.gov/30371261/.